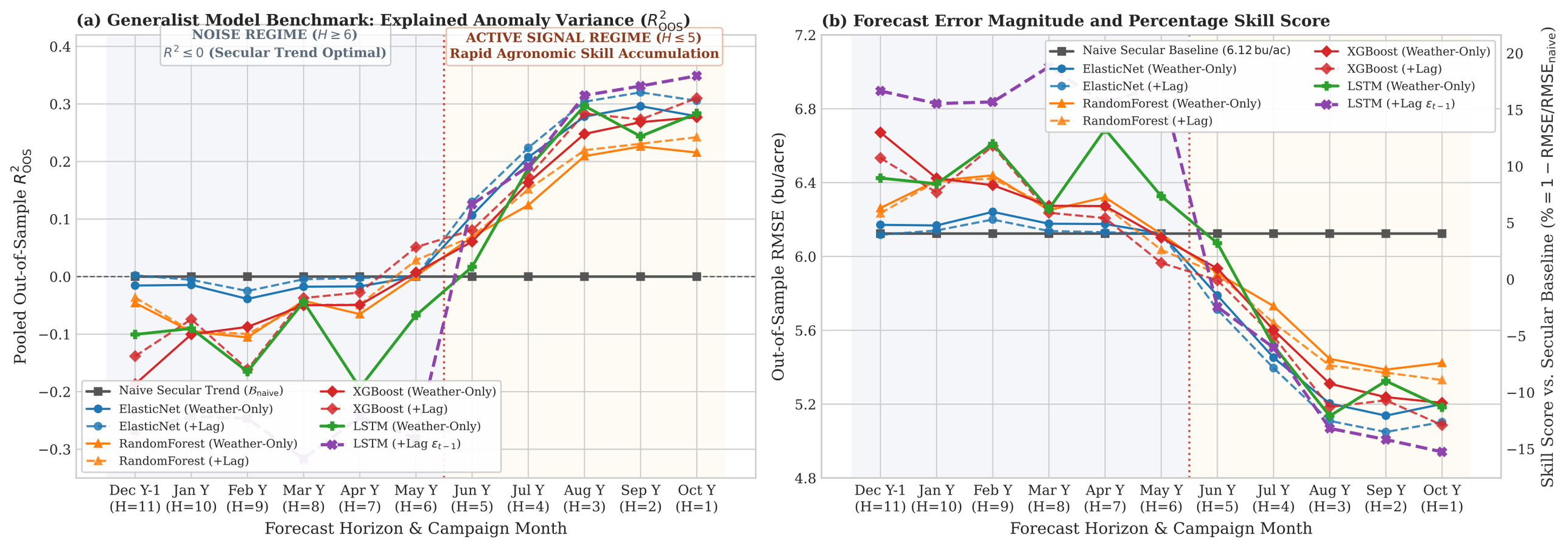

(a) Generalist Model Benchmark: Explained Anomaly Variance (R^2_{OOS})
NOISE REGIME ($H \geq 6$)
$R^2 \leq 0$ (Secular Trend Optimal)
ACTIVE SIGNAL REGIME ($H \leq 5$)
Rapid Agronomic Skill Accumulation
Pooled Out-of-Sample R^2_{OOS}
Naive Secular Trend ($\mathcal{B}_{naive}$)
ElasticNet (Weather-Only)
ElasticNet (+Lag)
RandomForest (Weather-Only)
RandomForest (+Lag)
XGBoost (Weather-Only)
XGBoost (+Lag)
LSTM (Weather-Only)
LSTM (+Lag ε_{t-1})
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Forecast Horizon & Campaign Month
(b) Forecast Error Magnitude and Percentage Skill Score
Out-of-Sample RMSE (bu/acre)
Skill Score vs. Secular Baseline (% = 1 − RMSE/RMSE$_{naive}$)
Naive Secular Baseline (6.12 bu/ac)
Forecast Horizon & Campaign Month